Closed meander numbers M35-M37

Evidence summary for the next proposed extension of OEIS A005315. As checked on 5 November 2026, the approved OEIS b-file ends at n=34. These three terms are not yet an approved OEIS update.

M35 **211433268000094602372146238438566**

time 526.72 min | peak sampled RSS 0.828 TB | matrix read/write 0/0 B

M36 **2358261838751729019212344104926012**

time 36.52 min | peak sampled RSS 0.946 TB | matrix read/write 0/0 B

M37 **263706529694495108792090969132427 92**

time 94.82 min | peak sampled RSS 1.771 TB | matrix read/write 0/0 B

What the result establishes

M37 completed all 66 transfer stages in 5,689.257 s (1 h 34 min 49 s) with 16 OpenMP threads. Its sampled RSS peak was 1,771,293,687,808 bytes, at stage 45. The heaviest numeric stage took 839.139 s. The program reported zero matrix-file reads and writes.

These are exact-integer computational outputs with internal bounds and small-order checks. No independent CRT or independent second algorithm has validated M35-M37, so the three proposed OEIS additions must be presented with that limitation.

Method, numerical checks and limitations

Algorithm

- The implementation represents the lower noncrossing matching by a deterministic weighted tree/centroid grammar. It propagates exact counts over upper/lower frontier geometry, with sparse support and cropped bitmap construction.
- Each transfer updates integer coefficient blocks. An exact source-maximum capacity is checked before numeric writes; overflow and resource guards stop a run instead of silently spilling matrices to disk. The final eight crossings are contracted by a small exact terminal recurrence.
- The M37 memory policy releases physical pages from dead translation and scratch buffers on Linux. It does not change the numeric kernels, the grammar, or the arithmetic bounds. This is a memory optimization, not a proved asymptotic improvement.

Validation evidence

- The frozen M37 source manifest covers 120 checked files. Linux self-tests passed 6,166 arithmetic checks, 19,414 brute-force matching-pair checks, and complete known orders 1-12 and 26. Four M32 full runs in an ABBA order agreed exactly and showed 15.97% lower median RSS with the full memory policy.
- The completed M37 record contains 66 sequential stage entries. Stage sums agree with the final result record, every stage records a checked arithmetic cap, and all stage matrix I/O counters are zero. Peak RSS was sampled every 5 ms. The raw JSON and stage log are included in the release candidate.
- M35, M36 and M37 were run with related but different versions of this tree implementation. Their runtimes are not a controlled same-order speed comparison. No formal proof of every code path or independent CRT for the new terms is claimed.

Reproduction and attribution

Frozen code and evidence

- The original M37 numerical source is preserved verbatim in source_frozen/. Run source_frozen/verify_bundle.py before installing dependencies or executing a job. The source manifest SHA-256 is f6835ac33bde4f40d28355e0b81e2ac2c7368da4fe1bcd491a1461e6d3b28dd5.
- Recorded M37 environment: Linux, Python 3.13.5, NumPy 2.3.5, Numba 0.68.0, psutil 7.2.2, 16 OpenMP threads, 1,900 GiB cgroup limit and 16 GiB reserve. The source bundle's historical setup_linux.sh is restricted to Python 3.12; the release candidate supplies a separate Python 3.13 setup wrapper without altering the frozen run source.
- The detailed seven-page Greek M37 report, the raw result and all 66 stage measurements are included under evidence/. The proposed OEIS b-file is under oeis/ and retains the official n=0..34 prefix exactly.

Related records and attribution

- OEIS A005315: https://oeis.org/A005315 . The current approved b-file ends at n=34 and already links the earlier C5/V2 software record.
- Earlier C5/V2 project: https://github.com/talepis/meander-checked-c5-v2 ; DOI 10.5281/zenodo.23687622. The tree/bitmap implementation is a distinct code lineage and merits its own repository and archival record.
- Vladimir Osipov, Exact enumeration of closed and open meanders in O*(2^n) time and space via a first-crossing partition of matching pairs, Zenodo DOI 10.5281/zenodo.22773575 (2026). The official A005315 entry credits his earlier reports of M29 and M30. This release does not assert priority for those terms.

Publication status: prepared locally for human scientific review. No new GitHub repository, Zenodo DOI or OEIS draft is asserted by this report.